CSE 4095 – Special Topics in Computer Engineering III: Introduction to Embedded Systems

Project # 2

(Due: 09.06.2023 23:59)

In this Project you will implement a C program to toggle the green LED (GPIO pin PA 5) once per 0.5 second, with a delay function driven by SysTick interrupts. Additionally, the frequency of the toggling should be increased to 1 second when the user presses the blue push button (GPIO pin PC 13). When the user presses the button again, the frequency should be reset to 0.5 second. Every time the user pushes the button, the frequency of toggling should change accordingly. You should use the EXTI interrupt to detect the push button signal.

In order to implement this project please implement the following functions:

- Implement **`void SysTick_Init()`** function to initialize system timer with 1000 ticks (i.e. 1 ms).

- Implement **`void SysTick_Handler(void)`** as interrupt service routine for SysTick interrupt. You can declare a global variable called **`TimeDelay`**. The handler should check if this variable is greater than 0. If so, decrement it.

- Implement **`void delay (void)`** function which will make busy waiting for the delay time length. Specifically, it should loop forever, waiting for **`TimeDelay`** to hit zero. Each time the interrupt handler is called, the value will decrease, so this delay routine will wait until the interrupt handler has been called **X** times depending on the delay is 1 second (**X** = 1000) or 0.5 second (**X** = 500).

- Implement **`void LED_Init(void)`** function to set up the green LED on GPIO PA5.

- Implement **`void EXTI_Init(void)`** function to configure blue push button on PC13 as input and enable external interrupt for this pin.

- Implement **`void EXTI15_10_IRQHandler(void)`** function to change the frequency of toggling.

- Implement **`int main(void)`** function that

    - invokes **`SysTick_Init(), LED_Init(),`** and **`EXTI_Init()`** functions first, and then,
    - have an infinite loop to toggle the green LED in every 0.5 seconds.

Important Notes:

- You can work in groups of 2-3 people.
- Please use Keil MDK-ARM IDE, create a project, enter and build the program successfully. Since we do not have a board we cannot observe the toggling LED or push buttons but building the project successfully is important.
- Please use the materials that we have covered in lectures.
- Please write necessary comments for the project.
- We use tools that automatically detect plagiarism among the submissions!
- In case of any form of copying and cheating on solutions, you will get **FF** grade from the course! You should submit your own work. In case of any forms of cheating or copying, both giver and receiver are equally culpable and suffer equal penalties.
- Please zip and submit your files using filename StudentNumbersProject2.zip (ex: 150713852_150713098_150101122_Project1.zip) to Canvas system (under Assignments tab).
- No late submission will be accepted.